

Hospital Admissions, Cardio-, Cerebro- and Peripheral Vascular Disease (ICD Codes 410, omitting 410.x2; 410-414; 426-427; 428; 429; 430-438; and 440-448)

In a single-pollutant model, the coefficient and standard error are estimated from a percent increase in risk of 0.65% (95% CI: 0.48-0.83%) for an increase of 10 $\mu\text{g}/\text{m}^3$ in same-day daily mean $\text{PM}_{2.5}$ concentrations (Bell et al. 2015, Table 1).

Hospital Admissions, Respiratory-2 (ICD Codes 490-492, 464-466, 480-487, 493)

In a single-pollutant model, the coefficient and standard error are estimated from a percent increase in risk of 0.25% (95% CI: 0.01-0.48%) for an increase of 10 $\mu\text{g}/\text{m}^3$ in same-day daily mean $\text{PM}_{2.5}$ concentrations (Bell et al. 2015, Table 1).

E.3.2 Ostro et al. (2009)

Ostro et al. (2009) estimated the association between ambient $\text{PM}_{2.5}$, EC, organic carbon (OC), NO_3 , and SO_4 on hospital admissions for respiratory diseases in children ages 5 to 19. The study used the California Office of Statewide Health Planning and Development, Healthcare Quality and Analysis Division hospitalization data from six California counties for the 2000 to 2003 study period. Ostro et al. (2009) classified hospital admissions into: all respiratory disease (ICD-9 codes 460-519), asthma (ICD-9 code 493), acute bronchitis (ICD-9 code 466), and pneumonia (ICD-9 codes 480-486). They aggregated the hospital admission data to the county level to create a daily time series of admissions for each county. Authors took air quality measurements from the California Air Resources Board, which captured speciated 24-hour average pollutant measurements using a filter-based Met One Speciation Air Sampling System. Meteorological measurements for average daily temperature and relative humidity came from the California Air Resources Board or the California Irrigation Management Information System. Authors analyzed data using a Poisson regression with time, day of the week, temperature, relative humidity, and pollutant as explanatory variables. Ostro et al. (2009) controlled for seasonality and time dependent effects by including a natural spline smoother for the daily time trend and meteorology.

Hospital Admissions, All Respiratory (ICD Codes 460-519)

In a single-pollutant model, the coefficient and standard error are estimated from an excess risk of 4.1% (95% CI: 1.8-6.4%) for a 14.6 $\mu\text{g}/\text{m}^3$ increase in the daily mean $\text{PM}_{2.5}$ concentrations, lagged by 3 days (Ostro et al. 2009, Table 2, pg. 477).

E.4 Emergency Room Visits

Table E-4 summarizes the health impact functions used to estimate the relationship between $\text{PM}_{2.5}$ and emergency room visits. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-4. Core Health Impact Functions for Particulate Matter and Emergency Room Visits

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Cardiovascular	Ostro et al.	2016	8 California cities	0-99		D24HourMean	0.000612	0.000422	Logistic	
Respiratory	Krall et al.	2013	Atlanta, GA	0-99		D24HourMean	0.000545	0.000267	Log-linear	
Respiratory	Krall et al.	2013	Birmingham, AL	0-99		D24HourMean	0.000968	0.000352	Log-linear	
Respiratory	Krall et al.	2013	St. Louis, MO	0-99		D24HourMean	0.000832	0.000329	Log-linear	
Respiratory	Krall et al.	2013	Dallas, TX	0-99		D24HourMean	0.001353	0.000588	Log-linear	

E.4.1 Ostro et al. (2016)

Ostro et al. (2016) investigated the association between short-term, source-specific (vehicular emissions, biomass burning, soil, and secondary NO_3 and SO_4 sources) $\text{PM}_{2.5}$ concentrations and emergency department visits for respiratory and cardiovascular diseases in eight cities in California from 2005 to 2008. Authors obtained medical and population subgroup data from the Office of Statewide Health Planning and Development in California, and diagnosis was defined with ICD-9 codes: all cardiovascular (390-459), ischemic heart disease (410-414), AMI (410), cardiac dysrhythmia (427), and heart failure (428). Ostro et al. (2016) conducted a case cross-over analysis, stratified by year and month, controlling for weather and day of the week covariates. Authors used a county-level logistic regression and random-effects meta-analysis to examine the association between source-specific $\text{PM}_{2.5}$ and emergency department visits for respiratory and cardiovascular diseases. Results indicate a positive association between vehicle $\text{PM}_{2.5}$ emissions and emergency department visits for all cardiovascular diseases.

ER Visits, All Cardiac Outcomes (ICD Codes 390-459)

In a single-pollutant model, the coefficient and standard error were estimated from the excess risk of 0.7% (95% CI: -0.2-1.7%) for a $11.4 \mu\text{g}/\text{m}^3$ (interquartile range) increase in daily mean $\text{PM}_{2.5}$ concentration, lagged by 2 days (Ostro et al. 2016, Table 4).

E.4.2 Krall et al. (2013)

Krall et al. (2013) investigated the associations between short-term, source-specific (traffic and coal combustion) ambient $\text{PM}_{2.5}$ exposure and emergency department visits for respiratory diseases in U.S. cities (Atlanta, GA, Birmingham, AL, St. Louis, MO, and Dallas, TX). Authors obtained medical data from hospital electronic billings for emergency department visits due to respiratory disease, identified using ICD-9 codes (460-465, 466, 477, 480-486, 491, 492, 493, 496, 786.07). Authors collected $\text{PM}_{2.5}$ concentrations from one ambient air monitor in each of the four cities and gathered meteorological data from the National Climatic Data Center. Krall et al. (2013) estimated source-specific $\text{PM}_{2.5}$ using apportionment models, which separate $\text{PM}_{2.5}$ sources based on chemical composition. This model also included data on gaseous

pollutant concentrations from the Community Multiscale Air Quality (CMAQ) with Tracers model. Krall et al. (2013) used Poisson time series regression models to analyze associations between short-term PM_{2.5} exposure and emergency department visits for respiratory diseases. They then compared source-specific PM_{2.5} exposures across cities to estimate associations with the emergency department visit data. To limit confounders, the authors adjusted models for indicator variables, meteorological variables, and long-term trends in emergency department visits.

ER Visits, Respiratory (ICD Codes 480-486, 491, 492, 496, 460-465, 466, 477, 493, 786.07)

In a single-pollutant model, the coefficient and standard error were estimated from a relative risk of 1.005 (95% CI: 1.000-1.010) for Atlanta, GA; 1.009 (95% CI: 1.003-1.015) for Birmingham, AL; 1.008 (95% CI: 1.002-1.014) for St. Louis, MO; and 1.012 (95% CI: 1.002-1.023) for Dallas, TX. All relative risks were calculated for a 9.16 µg/m³ increase in daily mean PM_{2.5} concentrations, lagged by 0 days (Krall et al. 2016, Figure 1).

E.5 Minor Effects

Table E-5 summarizes the health impacts functions used to estimate the relationship between PM_{2.5} and minor effects. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-5. Core Health Impact Functions for Particulate Matter and Minor Effects

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form
Work Loss Days	Ostro	1987	Nationwide	18-64		D24HourMean	0.004600	0.000360	Log-linear
Minor Restricted Activity Days	Ostro and Rothschild	1989	Nationwide	18-64	Ozone	D24HourMean	0.007410	0.000700	Log-linear
Hay Fever/Rhinitis	Parker et al.	2009	Nationwide	3-17	Summer O ₃ , PM _{2.5-10} , NO ₂ , SO ₂	Annual	0.025464	0.009618	Logistic

E.5.1 Ostro (1987)

Ostro (1987) estimated the impact of PM_{2.5} on the incidence of work-loss days (WLDs), restricted activity days (RADs), and respiratory-related RADs (RRADs) in a national sample of the adult working population, ages 18 to 65, living in metropolitan areas. The study population is based on the Health Interview Survey (HIS), conducted by the National Center for Health Statistics. The annual national survey results used in this analysis were conducted in 1976-1981. Ostro reported that two-week average PM_{2.5} levels were significantly linked to work-loss days, RADs, and RRADs, however there was some year-to-year variability in the results. Separate coefficients were developed for

each year in the analysis (1976-1981); these coefficients were pooled. The coefficient used in the concentration-response function presented here is a weighted average of the coefficients in Ostro (1987, Table III) using the inverse of the variance as the weight.

Work Loss Days

The coefficient used in the C-R function is a weighted average of the coefficients in Ostro (1987, Table III) using the inverse of the variance as the weight:

$$\beta = \frac{\left(\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2} \right)}{\left(\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2} \right)} = 0.0046$$

The standard error of the coefficient is calculated as follows, assuming that the estimated year-specific coefficients are independent:

$$\sigma_{\beta}^2 = \text{var} \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\sum_{i=1976}^{1981} \frac{1}{\sigma_{\beta_i}^2}} \right) = \text{var} \left(\frac{\sum_{i=1976}^{1981} \frac{\beta_i}{\sigma_{\beta_i}^2}}{\gamma} \right) = \sum_{i=1976}^{1981} \text{var} \left(\frac{\beta_i}{\sigma_{\beta_i}^2 \times \gamma} \right)$$

This eventually reduces down to:

$$\sigma_{\beta}^2 = \frac{1}{\gamma} \Rightarrow \sigma_{\beta} = \sqrt{\frac{1}{\gamma}} = 0.00036$$

Incidence Rate: daily work-loss-day incidence rate per person ages 18 to 64 = 0.00595 (U.S. Bureau of the Census, 1997, No. 22; Adams et al., 1999, Table 41)

Population: adult population ages 18 to 64

E.5.2 Ostro and Rothschild (1989)

Ostro and Rothschild (1989) estimated the impact of PM_{2.5} and ozone on the incidence of minor restricted activity days (MRADs) and respiratory-related restricted activity days (RRADs) in a national sample of the adult working population, ages 18 to 65, living in metropolitan areas. The study population is based on the Health Interview Survey (HIS), conducted by the National Center for Health Statistics. In publications from this ongoing survey, non-elderly adult populations are generally reported as ages 18-64. From the study, it is not clear if the age range stops at 65 or includes 65 year olds. We apply the C-R function to individuals ages 18-64 for consistency with other studies estimating impacts to non-elderly adult populations. The annual national survey results